



Department of Electrical and Computer
Engineering
Instructor : Hassan Tfaily

Lab 3 : “DIP Switches”

Introduction and objectives

In this lab, you will be introduced to the switches connected to the MC9S12DT256 microcontroller. Switches are the simplest and most commonly used input components in embedded systems.

You will learn how to read the status of a switch when it is pressed or released, since it will change the logic level on the pin it is connected to, thus allowing the microcontroller to detect user input and respond accordingly.

This lab session is primarily focused on writing assembly code to configure the pins connected to the switches using the appropriate data direction register, reading the switch states and performing logic decisions based on their values.

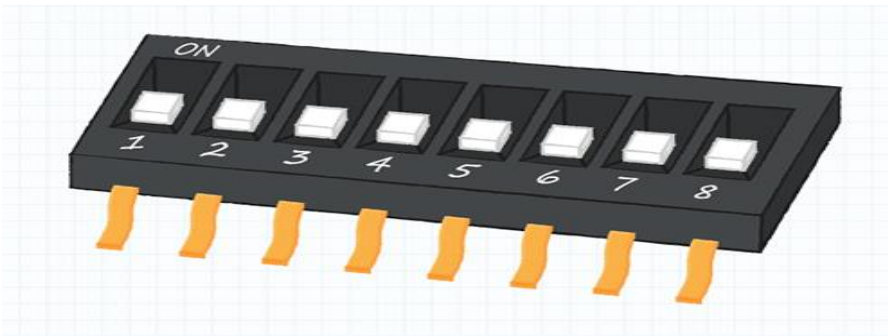
Pre Lab3:

In this preliminary task, you are required to answer the following questions :

Ex1:

In **Lab1**, you learned that **LEDs** are **OUTPUT** devices that emit light (ON), when charged with a current, they may be active LOW or active HIGH.

DIP switches are **INPUT** , made up of a set of switches in a single unit, the switches position is to be set manually.



- a) Locate the DIP switch in the schematic diagram, follow its connection, identify the PORT it is connected to.
- b) As you did in Lab1, configure the register that indicates the direction of the pins connected to the DIP switch.
- c) Are the DIP switches connected as active LOW or active HIGH?
- d) Switches are typically connected in either pull-up or pull-down configuration:
 - i) In a pull-up configuration, the pin reads a high logic level '1' when the switch is open and a low logic level '0' when it is pressed.
 - ii) In a pull-down configuration, the logic is reversed.

Check the schematic diagram 'CSMB12_SCH_A1_0' and deduce the configuration of the switches.

- e) **PUCR** (pull up control register) is used to enable or disable INTERNAL PULL-UP resistors for certain pins of the MCU.
It is assigned on a PORT-BASIS, it selects pullup resistors for ALL THE PINS IN THIS PORT that are configured as inputs.

By referring to the "MC9S12XS256RMV1" datasheet (located in Microcontroller folder), study the configuration of the PUCR register: which PIN of the PUCR must be enabled to activate the pullup resistors on the input pins PB0→PB3 of the PORTB, and what will be the configuration of the PUCR?

- f) *Refer to the "instruction_set" datasheet (located in folder Instruction), read carefully and understand the program below :*

```

Counter1 dc.b 3

        ldaa $1010
        bita #$01
        beq Label1
again    lsla
        dec Counter1
        bne again
Label1   staa $1001

```

- i. Write the byte stored in the address \$1001, when :
 - The byte \$01 is stored in address \$1010
 - The byte \$04 is stored in address \$1010
- ii. Replace both instruction “ldaa”, “bita” and “beq” with :
brclr \$1010,\$01,Label1 and make the appropriate modifications.

To create variables or constants in your code, use DC and DS:

```

Counter1 DC.B 7
Counter2 DS.B 1
Counter3 DC.W 4

```

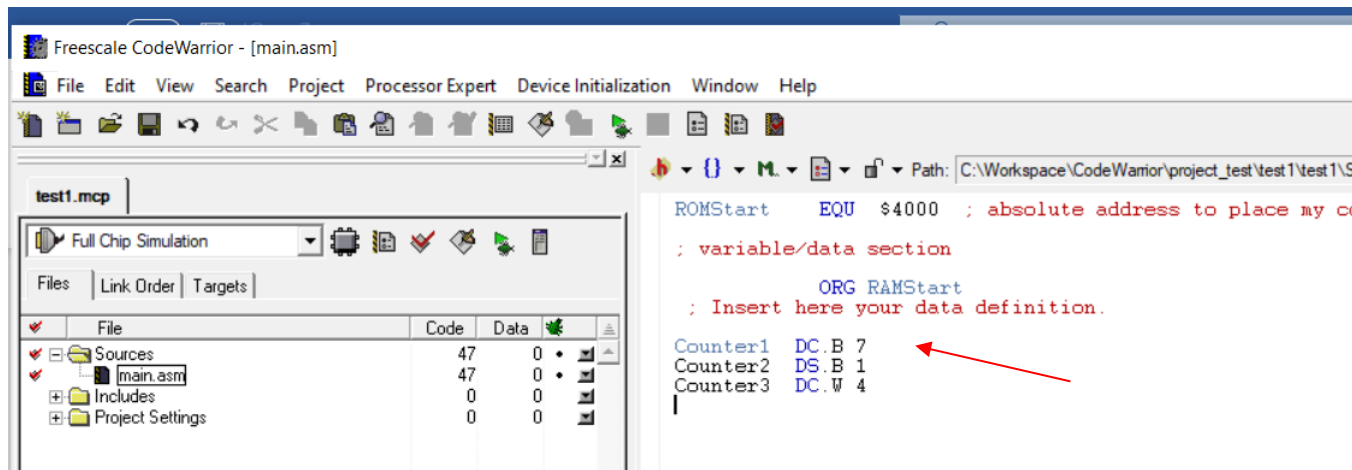
DC will allocate and initialize a value for the variable, while DS only allocate bytes without initialization.

Use .B for Bytes, .W for words (2 bytes).

For example : DC allocates one byte for Counter1 and initializes to 7.

DS allocates 1 byte for Counter2 without initialization.

DC allocates one word for Counter3 and initializes it to 4.



Lab 3 : “DIP Switches”

Write fully commented programs for the PBMCUSLK board, including appropriate labels, memory operands and constants, to solve the 4 tasks below:

Task1:

The first program should do the following:

Blink the LEDs continuously with a blinking time of **100ms** when :
DIP1 and DIP3 are pressed and DIP2 and DIP4 are not pressed.

Task 2 :

The second program should do the following:

Shift the light to the left, from LED1 to LED4 continuously with a time delay of **100ms** if DIP1 is pressed, and turn OFF the LEDs once DIP4 is pressed.

Task 3:

The third program should do the following:

Turn ON , *continuously* , an LED when the corresponding switch is pressed.

For example:

If DIP1 is pressed → LED1 will turn ON , *the other LEDs are OFF*.

If DIP1 , DIP2 are pressed → LED1 , LED2 will turn ON , *the other LEDs are OFF*.

Task4:

The fourth program should do the following:

use the LSBs of SW3 (DIP1,DIP2,DIP3) as a multiplier of the blinking time delay of 100ms.

For example:

when DIP3 DIP2 DIP1 = OFF OFF ON the blinking time is 100ms , the LEDs turn ON for 1*100 ms and then turn OFF.

when DIP3 DIP2 DIP1 = ON ON ON the blinking time is 7*100ms, the LEDs turn ON for 7*100ms and then turn OFF.

N.B: A switch is 'ON' means : this switch is pressed

A switch is OFF' means : this switch is not pressed .
